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WO-100-05 — Replace VALVE, GATE, WEDGE, OS&Y 2" P150 (2"-P150-SC-3010/A2)

Area	A100 Tank farm	**Task**	VR Valve replacement in kind
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	T-102 — T-102 tank 12' dia × 22' high — E 44'-0" N 26'-6" EL 100'-6"		
Equipment	T-102	**Lines**	2"-P150-SC-3010
Isometrics	ISO-SC-3010 (owner's drawings, attached)		
Permits	General work permit		

Reason

Condensate isolation valve passing; replace in kind.

Scope of work

1. Isolate per the isolation list, drain and vent 2"-P150-SC-3010 between the isolations.
2. Unbolt and remove VALVE, GATE, WEDGE, OS&Y 2" P150 (2"-P150-SC-3010/A2) at E 37'-5/8" N 26'-6" EL 104'-6".
3. Inspect both flange faces (SW-NDE-5); dress or report damage.
4. Install a new VALVE, GATE, WEDGE, OS&Y 2" P150 of the same size, class and face-to-face (GATE2-P150), with new gaskets and studs, stem orientation as removed.
5. Tighten the studs to the stated preload in three passes, cross pattern (SW-WO-5).
6. Remove blinds, restore the line-up, leak test the two joints on start-up (SW-TEST-2).

Affected items: 2"-P150-SC-3010/A2, 2"-P150-SC-3010/A1, 2"-P150-SC-3010/A4

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-SC-3010	2"-P150-SC-3010	2"-P150-SC-3010	1	0 error(s), 1 warning(s) — see below

ISO-SC-3010 (PDF): ../isometrics-imported/ISO-SC-3010.pdf — the sheets that hold the work follow the ITP.

Drawing	Sheet	Level	Finding
ISO-SC-3010	1	warning	FROM label 'T-102 NOZ. N8' is not on the drawing

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
T-102	equipment	T-102 tank 12' dia × 22' high		E 44'-0" N 26'-6" EL 100'-6"
2"-P150-SC-3010/A2	item	VALVE, GATE, WEDGE, OS&Y 2" P150	2"-P150-SC-3010	E 37'-5/8" N 26'-6" EL 104'-6"
2"-P150-SC-3010/A1	item	GASKET 2" P150 COMPRESSED SHEET	2"-P150-SC-3010	E 37'-6-1/16" N 26'-6" EL 104'-6"
2"-P150-SC-3010/A4	item	GASKET 2" P150 COMPRESSED SHEET	2"-P150-SC-3010	E 36'-7-3/16" N 26'-6" EL 104'-6"
T-102-N8	nozzle	Nozzle T-102-N8 (condensate out, 2" P150) on T-102		E 37'-6-1/8" N 26'-6" EL 104'-6"
2"-P150-SC-3010/B3	item	VALVE, GATE, WEDGE, OS&Y 2" P150	2"-P150-SC-3010	E 34'-0" N -2'-0" EL 123'-6-3/16"
2"-P150-SC-3010/B6	item	FLANGE WELD-NECK 2" P150 RAISED FACE	2"-P150-SC-3010	E 34'-0" N -2'-0" EL 124'-1-13/16"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	T-102-N8	2"-P150-SC-3010	Line end at nozzle T-102-N8: the equipment is out of service, drained and isolated under the shutdown isolation			
2	valve	2"-P150-SC-3010/B3	2"-P150-SC-3010	Close and lock VALVE, GATE, WEDGE, OS&Y 2" P150 (2"-P150-SC-3010/B3)	E 34'-0" N -2'-0" EL 123'-6-3/16"		
3	blind	2"-P150-SC-3010/B6	2"-P150-SC-3010	Fit a line blind between 2"-P150-SC-3010/B3 and FLANGE WELD-NECK 2" P150 RAISED FACE (2"-P150-SC-3010/B6), work side	E 34'-0" N -2'-0" EL 123'-11-11/16"		

Parts

Description	Qty	Replaces
VALVE, GATE, WEDGE, OS&Y 2" P150	1	2"-P150-SC-3010/A2

Description	Qty	Replaces
GASKET 2" P150 COMPRESSED SHEET	1	2"-P150-SC-3010/A1
STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P150	1	2"-P150-SC-3010/A3
GASKET 2" P150 COMPRESSED SHEET	1	2"-P150-SC-3010/A4
STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P150	1	2"-P150-SC-3010/A6

Inspections

Id	Inspection	Criteria	Basis
WO-100-05-I1	Flange faces of both joints before the new gaskets go in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
2"-P150-SC-3010/A1	4 × 0.625" studs (2"-P150-SC-3010/A3)	13,806	3	
2"-P150-SC-3010/A4	4 × 0.625" studs (2"-P150-SC-3010/A6)	13,806	3	

Test — Service leak test

Description	Both remade joints checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

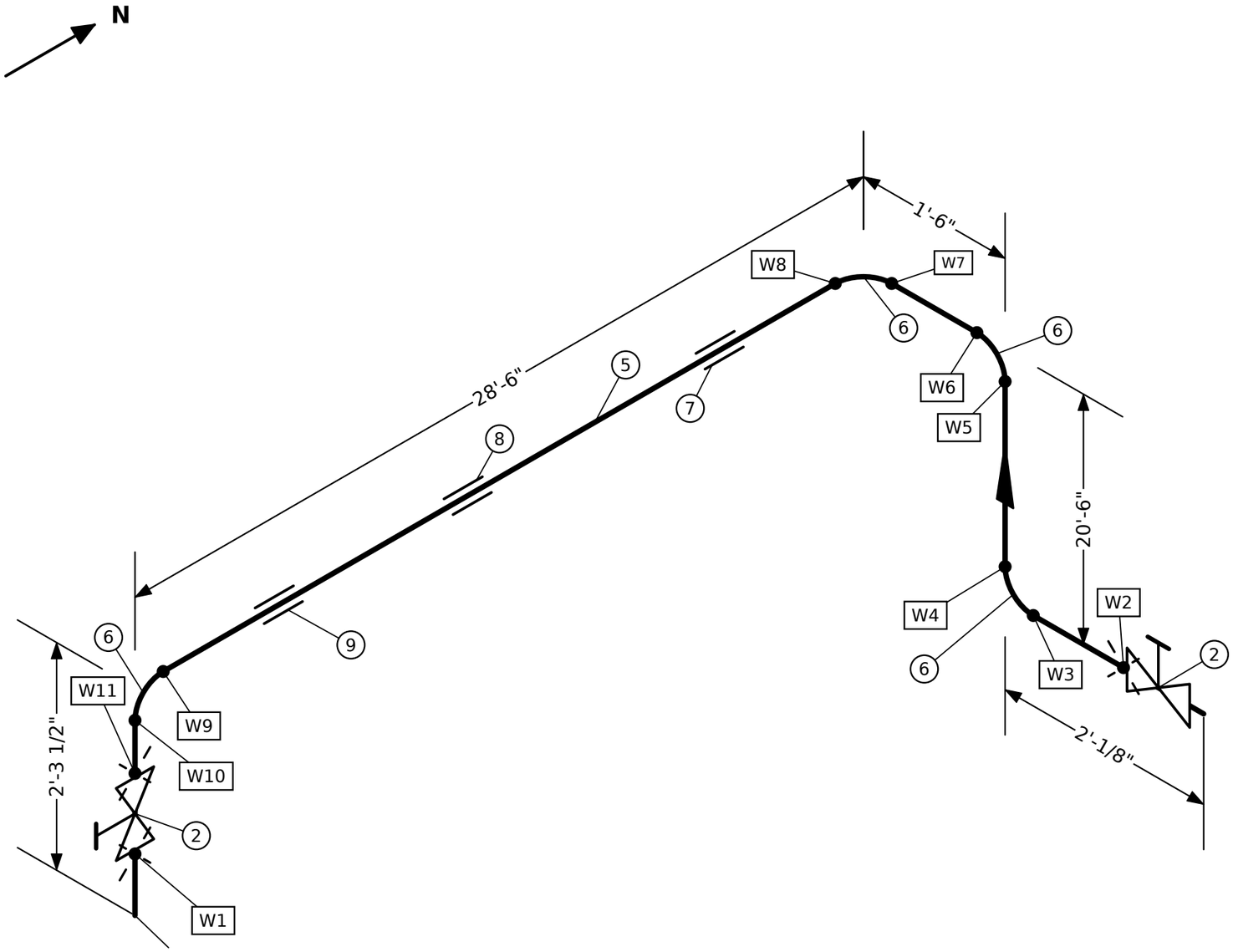
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Valve 2"-P150-SC-3010/A2 removed; flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New valve verified: type, size, class, face-to-face and tag	GATE2-P150	Matches the parts list and the line class	Material receiving record	R	R	—				
4	New gaskets fitted, studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated; three passes, cross pattern	Bolt-up record per joint	W	W	—				
5	Service leak test of the remade joints on start-up	SW-TEST-2	No leak at either joint	Leak test record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
2"-P150-SC-3010/A2	affected	ISO-SC-3010	1	
2"-P150-SC-3010/A1	affected	ISO-SC-3010	1	
2"-P150-SC-3010/A4	affected	ISO-SC-3010	1	
2"-P150-SC-3010/B3	isolation: valve	ISO-SC-3010	1	
2"-P150-SC-3010/B6	isolation: blind	ISO-SC-3010	1	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	4	2"	GASKET 2" P150 COMPRESSED SHEET
2	2	2"	VALVE, GATE, WEDGE, OS&Y 2" P150
3	4	2"	STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P150
4	3	2"	FLANGE WELD-NECK 2" P150 RAISED FACE
5	49'-8 3/8"	2"	PIPE 2" P150 BEVELLED ENDS
6	4	2"	ELBOW 90° R = 5/3 OD 2" P150
7	1	2"	HANGER FOR 2" LINE (HANGER-2x78.1)
8	1	2"	HANGER FOR 2" LINE (HANGER-2x88.0)
9	1	2"	HANGER FOR 2" LINE (HANGER-2x93.0)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

STEAM CONDENSATE

NOTES:
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	YES	-
PWHT	NDE	FLUID
-	-	SC

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
2"-P150-SC-3010	C150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-SC-3010	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

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